

Elephants: When is Enough, Enough?

“Ultimately, if a habitat is undesirably changed by elephants, then their removal should be considered �even by culling.” National Geographic (Earth Almanac)�December 1999

A large National Park in South Africa contains approximately 11,000 elephants. Management A large National Park in South Africa contains approximately 11,000 elephants. Management policy requires a healthy environment that can maintain a stable herd of 11,000 elephants. Each year park rangers count the elephant population. During the past 20 years whole herds tion. During the past 20 years whole herds have been removed to keep the population as close to 11,000 as possible. This process involved shooting (for the most part) and occasionally relocating approximately 600 to 800 elephants per year

Recently, there has been a public outcry against the shooting of these elephants. In addition, there has been a public outcry against the shooting of these elephants. In addition, it is no longer feasible to relocate even a small population of elephants each year. A contraceptive dart, how-ever, has been developed that can prevent a mature elephant cow from conceiving for a period of two years.

Here is some information about the elephants in the Park:

There is very little emigration or immigration of elephants. The gender ratio is very close to 1:1 and control measures have endeavored to maintain parity. The gender ratio of newborn calves is also about 1:1. Twins are born about 1.35% of the time. Cows first conceive between the ages of 10 and 12 and produce, on average,

a calf every 3.5 years until they reach an age of about 60. Gestation is approximately 22 months. The contraceptive dart causes an elephant cow to come into oestrus every month (but not conceiving). Elephants usually have courtship only once in 3.5 years, so the monthly cycle can cause additional stress. A cow can be darted every year without additional detrimental effects. A mature elephant cow will not be able to conceive for 2 years after the last darting. Between 70% and 80% of newborn calves survive to age 1 year. Thereafter, the survival rate is uniform across all ages and is very high (over 95%), until about age 60; after age 60, the survival rate is very high (over 95%), until about age 70. It is a good assumption that elephants die before reaching age 70. There is no hunting and negligible poaching in the Park.

The park management has a rough data file of the approximate ages and gender of the elephants they have transported out of the region during the past 2 years. This data is available on website: <http://www.comap.com/icm/icm2000data.xls>. Unfortunately no data is available for the elephants that have been shot or remain in the Park. Your overall task is to develop and use model to investigate how the contraceptive dart might be used for to investigate how the contraceptive dart might be used for

population control. Specifically

Task 1: Develop and use a model to speculate about the likely survival rate for elephants. Task 1: Develop and use a model to speculate about the likely survival rate for elephants aged 2 to 60. Also speculate about the current age structure of the elephant population.

Task 2 : Estimate how many cows would need to be darted each year to keep the population fixed at approximately 11,000 elephants. Show how the uncertainty in the data at your disposal affects your estimate. Comment on any changes in the age structure of the population and how this might affect tourists. (You may want to look ahead about 30-60 years.)

Task3: If it were feasible to relocate between 50 and 300 elephants per year, how would this reduce the number of elephants to be darted? Comment on the trade-off between darting and relocation.

Task 4 : Some opponents of darting argue that if there were a sudden loss of a large number of elephants (due to disease or uncontrolled poaching), even if darting stopped immediately, the ability of the population to grow again would be seriously impeded. Investigate and respond to this concern.

Task 5: The management in the Park is skeptical about modeling. In particular, they argue that a lack of complete data makes a mockery of any attempt to use models to guide their decisions. In addition to your technical report, include a carefully crafted report (3-page maximum) written explicitly for the park management that responds to their concerns and provides advice. Also suggest

ways to increase the park managers confidence in your model and in your conclusions.

Task 6: If your model works, other elephant parks in Africa would be interested in using it.

Prepare a darting plan for parks of various sizes (300�25,000 elephants), with slightly different survival rates and transportation possibilities.

大象群落的兴衰

归根到底，如果象群对于栖息地造成不尽人意的影响，就要考虑对它们的驱除，即使是运用淘汰法则。国家地理杂志（地球年鉴）1999 年 12 月

在位于南非的一个巨大的国家公园里，栖息着近乎 11000 只象。管理策略要求一个健康的环境以便维持 11000 只象的稳定群落。公园的管理员们逐年统计象的总数。在过去的 20 年间，整个群落经受驱除得以保持其总数尽量接近 11000 只。这个过程涉及枪杀（对于大部分）和每年转移近乎 600 到 800 只象到异地。

近年来，公众抗议枪杀这些象。此外，即使每年转移少量的象也是不可能了。然而，一种避孕注射法开发成功，它可以在两年期间内阻止一只成熟的母象受孕。

下面是一些关于这个公园内象的信息：

很少发生象本身移入移出该公园的事。

性别比非常接近 1:1，而且采取控制措施力求维持均衡。

新生幼象的性别比也是 1:1 左右。双胞胎的机会接近于 1.35%。

母象在 10 岁和 12 岁之间第一次怀孕，平均每 3.5 年产下一个崽儿，直到 60 岁左右为止。

怀孕期约为 22 个月。

避孕注射使一只母象每个月发情（但不怀孕）。象通常在 3.5 年内仅求偶一次，所以，上述按月周期能够引起附加的反应。

一只母象可以每年注射而没有任何有害的影响。一只成熟的母象在上次注射后两年内将不能怀孕。

新生幼象中的 70%到 80%活到一岁，其后，存活率非常高（超过 95%）并且在各年龄段一致，直到 60 岁左右；假定象死于 70 岁之前是恰当的。在这个公园内没有狩猎，偷猎也是微乎其微。

公园管理部门有一个粗略的数据文件，其中列出近两年内由这个地区运出的象的大致年龄和性别。这组数据可在网站 <http://www.comap.com/icm/icm2000data.xls> 上找到。可惜的是，没有关于在这个公园内被射杀和留下来的象的可用数据。你的全部任务是发展和利用模型来研究避孕注射会如何用于控制象的数量。特别是：

任务 1：发展和利用一个模型来推测年龄在 2 岁到 60 岁之间象的合理存活率。并且推测这个大象群落的当前年龄结构。

任务 2：估计每年有多少只母象需要避孕注射以保持这个群落固定在 11000 只象左右。说明被处理数据的不确定性如何影响你的估计。试加评论这个群落年龄结构的任何改变以及会如何影响旅游者。（你或许要前瞻 30-60 年左右。）

任务 3：假如每年转移 50 至 300 只象是可行的，这会怎样减少承受避孕注射的象只数量？试加评定避孕注射和转移之间的折衷办法。

任务 4：若干反对避孕注射的人提出疑问，如果发生一场大量象只的突然灭绝（由于疾病或不受控制的偷猎），即使立即停止避孕注射，这个群落重新壮大的能力也会受到严重阻碍。对这个顾虑进行研究并作出回应。

任务 5：这个公园的管理部门不相信建模。他们特别表示，由于缺少完整的数据，任何通过模型来引导他们作出决定的尝试都构成一种愚弄。除了你的技术报告之外，请附上一份字斟句酌写给公园管理部门的报告（最多三页），对于他们的疑虑作出回应并且给予劝告。还要提出一些办法来增加公园管理部门对于你的模型和结论的信赖程度。

任务 6：如果你的模型有效，南非的其他大象公园会乐于采用它。请为各种规模的公园（300 至 25000 只象）准备一项避孕注射计划，同时带有略微不同的存活率和转运可能性。